Panasonic
INDUSTRY

Data Sheet

High Speed, Low Loss Multi-layer Materials

Low Dk glass cloth version

MEGTRON6

Laminate R-5775(N)
Prepreg R-5670(N)

Apr. 2022 No.22040130

Partnering to go beyond.

Electronic Materials
Panasonic Industry

Specification / Laminate R-5775(N)

Property			Units	Test Method	Condition	Typical Value
THERMAL	Glass Transition Temp (Tg)		C	DSC	As received	185
				DMA	As received	210
	Thermal Decomposition Temp (Td)		C	TGA	As received	410
	Time to Delam (T288)	Without Cu	Min	IPC TM-650 2.4.24.1	As received	> 120
		With Cu	Min	IPC TM-650 2.4.24.1	As received	> 120
	CTE : α1	X - axis	ppm / C	IPC TM-650 2.4.24	< Tg	14 - 16
		Y - axis	ppm / C	IPC TM-650 2.4.24	< Tg	14 - 16
		Z - axis	ppm / C	IPC TM-650 2.4.24	< Tg	45
	CTE : α2	Z - axis	ppm / C	IPC TM-650 2.4.24	> Tg	260
ELECTRICAL	Volume Resistivity		MΩ - cm	IPC TM-650 2.5.17.1	C-96/35/90	1×10^{9}
	Surface Resistivity		MΩ	IPC TM-650 2.5.17.1	C-96/35/90	1×10^{8}
	Dielectric Constant (Dk)	@ 1GHz	-	IPC TM-650 2.5.5.9	C-24/23/50	3.40
		@ 13GHz	-	*Note 1	C-24/23/50	3.34
	Dissipation Factor (Df)	@ 1GHz	-	IPC TM-650 2.5.5.9	C-24/23/50	0.002
		@ 13GHz	-	*Note 1	C-24/23/50	0.0037
PHYSICAL	Water Absorption		%	IPC TM-650 2.6.2.1	D-24/23	0.14
	Peel Strength	1oz (H-VLP)	kN / m	IPC TM-650 2.4.8	As received	0.8
	Flammability		-	UL	C-48/23/50	94V-0

Sample thickness : 29.5 mil = 0.750 mm (Core Type 30)

Note 1 : Balanced-type Circular Disk Resonance Method [IEC 63185 (2020)]

* The data in the above table are not guaranteed values.

Specification / Laminate R-5775(N)

1GHz ; IPC TM650-2.5.5.9
13-58GHz ; Balanced-type Circular Disk Resonance Method [IEC 63185 (2020)]

Core Type	Actual Thickness		Cloth Style	ply	Typical Resin Content (%)	Typical Dk					
	mil	mm				1 GHz	13 GHz	24 GHz	36 GHz	47 GHz	58 GHz
2	2.0	0.050	1035	1	67	3.25	3.19	3.19	3.19	3.19	3.19
2.6	2.6	0.065	1078	1	59	3.37	3.29	3.29	3.29	3.29	3.29
3	3.0	0.075	1078	1	65	3.28	3.22	3.22	3.22	3.22	3.22
3.5	3.5	0.090	1078	1	70	3.22	3.16	3.16	3.16	3.16	3.16
4	3.9	0.100	2013	1	56	3.40	3.34	3.34	3.34	3.34	3.34
4	3.9	0.100	1035	2	67	3.25	3.19	3.19	3.19	3.19	3.19
5	5.0	0.127	1078	2	59	3.37	3.29	3.29	3.29	3.29	3.29
5	4.9	0.125	2116	1	56	3.40	3.34	3.34	3.34	3.34	3.34
6	5.7	0.146	1078	2	65	3.28	3.22	3.22	3.22	3.22	3.22
7	7.0	0.178	1078	2	70	3.22	3.16	3.16	3.16	3.16	3.16
8	7.9	0.200	2013	2	56	3.40	3.34	3.34	3.34	3.34	3.34
10	9.8	0.250	2116	2	56	3.40	3.34	3.34	3.34	3.34	3.34
12	11.8	0.300	2013	3	56	3.40	3.34	3.34	3.34	3.34	3.34
16	15.7	0.400	2013	4	56	3.40	3.34	3.34	3.34	3.34	3.34
20	19.7	0.500	2116	4	56	3.40	3.34	3.34	3.34	3.34	3.34
25	24.6	0.625	2116	5	56	3.40	3.34	3.34	3.34	3.34	3.34
30	29.5	0.750	2116	6	56	3.40	3.34	3.34	3.34	3.34	3.34

* The data in the above table are not guaranteed values.

Specification / Laminate R-5775(N)

1GHz ; IPC TM650-2.5.5.9
13-58GHz ; Balanced-type Circular Disk Resonance Method [IEC 63185 (2020)]

Core Type	Actual Thickness		Cloth Style	ply	Typical Resin Content (%)	Typical Df					
	mil	mm				1 GHz	13 GHz	24 GHz	36 GHz	47 GHz	58 GHz
2	2.0	0.050	1035	1	67	0.002	0.0037	0.0040	0.0043	0.0044	0.0046
2.6	2.6	0.065	1078	1	59	0.002	0.0037	0.0040	0.0042	0.0044	0.0046
3	3.0	0.075	1078	1	65	0.002	0.0037	0.0040	0.0042	0.0044	0.0046
3.5	3.5	0.090	1078	1	70	0.002	0.0037	0.0040	0.0043	0.0044	0.0046
4	3.9	0.100	2013	1	56	0.002	0.0037	0.0040	0.0042	0.0044	0.0046
4	3.9	0.100	1035	2	67	0.002	0.0037	0.0040	0.0043	0.0044	0.0046
5	5.0	0.127	1078	2	59	0.002	0.0037	0.0040	0.0042	0.0044	0.0046
5	4.9	0.125	2116	1	56	0.002	0.0037	0.0040	0.0042	0.0044	0.0046
6	5.7	0.146	1078	2	65	0.002	0.0037	0.0040	0.0042	0.0044	0.0046
7	7.0	0.178	1078	2	70	0.002	0.0037	0.0040	0.0043	0.0044	0.0046
8	7.9	0.200	2013	2	56	0.002	0.0037	0.0040	0.0042	0.0044	0.0046
10	9.8	0.250	2116	2	56	0.002	0.0037	0.0040	0.0042	0.0044	0.0046
12	11.8	0.300	2013	3	56	0.002	0.0037	0.0040	0.0042	0.0044	0.0046
16	15.7	0.400	2013	4	56	0.002	0.0037	0.0040	0.0042	0.0044	0.0046
20	19.7	0.500	2116	4	56	0.002	0.0037	0.0040	0.0042	0.0044	0.0046
25	24.6	0.625	2116	5	56	0.002	0.0037	0.0040	0.0042	0.0044	0.0046
30	29.5	0.750	2116	6	56	0.002	0.0037	0.0040	0.0042	0.0044	0.0046

* The data in the above table are not guaranteed values.

Specification / Prepreg R-5670(N)

1GHz ; IPC TM650-2.5.5.9
13-58GHz ; Balanced-type Circular Disk Resonance Method [IEC 63185 (2020)]

Cloth Style	Resin Content (%)	Typical Thickness (um)	Typical Dk					
			1 GHz	13 GHz	24 GHz	36 GHz	47 GHz	58 GHz
1035	72	60	3.20	3.13	3.13	3.13	3.13	3.13
	75	68	3.16	3.10	3.10	3.10	3.10	3.10
	77	74	3.13	3.08	3.08	3.08	3.08	3.08
1078	66	77	3.26	3.20	3.20	3.20	3.20	3.20
	70	89	3.22	3.16	3.16	3.16	3.16	3.16
	74	104	3.17	3.11	3.11	3.11	3.11	3.11
	77	118	3.13	3.08	3.08	3.08	3.08	3.08
2013	56	98	3.40	3.34	3.34	3.34	3.34	3.34
	59	106	3.37	3.29	3.29	3.29	3.29	3.29
2116	56	125	3.40	3.34	3.34	3.34	3.34	3.34
	58	132	3.37	3.31	3.31	3.31	3.31	3.31

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Specification / Prepreg R-5670(N)

1GHz ; IPC TM650-2.5.5.9
13-58GHz ; Balanced-type Circular Disk Resonance Method [IEC 63185 (2020)]

Cloth Style	Resin Content (%)	Typical Thickness (um)	Typical Df					
			1 GHz	13 GHz	24 GHz	36 GHz	47 GHz	58 GHz
1035	72	60	0.002	0.0037	0.0040	0.0043	0.0044	0.0046
	75	68	0.002	0.0037	0.0040	0.0043	0.0044	0.0046
	77	74	0.002	0.0037	0.0040	0.0043	0.0044	0.0046
1078	66	77	0.002	0.0037	0.0040	0.0043	0.0044	0.0046
	70	89	0.002	0.0037	0.0040	0.0043	0.0044	0.0046
	74	104	0.002	0.0037	0.0040	0.0043	0.0044	0.0046
	77	118	0.002	0.0037	0.0040	0.0043	0.0044	0.0046
2013	56	98	0.002	0.0037	0.0040	0.0042	0.0044	0.0046
	59	106	0.002	0.0037	0.0040	0.0042	0.0044	0.0046
2116	56	125	0.002	0.0037	0.0040	0.0042	0.0044	0.0046
	58	132	0.002	0.0037	0.0040	0.0042	0.0044	0.0046

* The data in the above table are not guaranteed values.

++ Before purchase ++

【Notes before you use】

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